Pattern Classification of Human Activities Using Smartphone Sensor Data

Ananya Chittineni

1 Introduction

Human Activity Recognition (HAR) is an important problem in pattern classification and machine learning. The objective of HAR is to automatically identify human physical activities using sensor data collected from smartphones and wearable devices. Modern smartphones contain sensors such as accelerometers and gyroscopes that continuously record motion signals. These signals can be processed and classified to recognize different activities performed by a user.

HAR has several real-world applications. In healthcare, it can be used to monitor patient mobility and detect abnormal activity patterns. In fitness tracking, it can be used to monitor exercise routines and daily physical activity. In smart home systems, HAR can help understand user behavior and support context-aware applications. However, classifying human activities is challenging because some activities produce very similar sensor patterns. For example, sitting and standing are both relatively static activities, while walking, walking upstairs, and walking downstairs can share similar periodic motion patterns.

In this project, human activity recognition is formulated as a multi-class pattern classification problem. Two datasets are used: the UCI Human Activity Recognition Using Smartphones dataset and the WISDM dataset. The UCI HAR dataset contains engineered time-domain and frequency-domain features extracted from smartphone accelerometer and gyroscope signals. In contrast, the WISDM dataset contains raw accelerometer readings, making it noisier and more challenging. Using both datasets allows the project to compare classification performance on engineered features versus raw sensor-derived features.

Several classification techniques are evaluated, including k-Nearest Neighbors (kNN), Support Vector Machines (SVM), Random Forest, Gaussian Naive Bayes, a Multivariate Gaussian MAP classifier, and a non-parametric Parzen/KDE classifier. In addition, Principal Component Analysis (PCA) and Sequential Forward Floating Selection (SFFS) are applied to study the effect of dimensionality reduction and feature selection. The models are evaluated using classification accuracy, confusion matrices, and stability analysis over multiple train-test splits.

2 Dataset Description

2.1 UCI HAR Dataset

The UCI HAR dataset contains smartphone sensor data collected from 30 participants performing six daily activities: walking, walking upstairs, walking downstairs, sitting, standing, and laying. The dataset contains 10,299 total samples, where each sample represents a time window of sensor measurements. Each sample is represented using 561 numerical features extracted from accelerometer and gyroscope signals.

The features include statistical and signal-processing measurements such as mean, standard deviation, signal magnitude area, energy, and frequency-domain coefficients. Since the dataset already contains engineered features, it provides a structured high-dimensional representation of human motion. This makes it suitable for comparing classification algorithms and dimensionality reduction methods.

The six activity classes in the UCI HAR dataset are:

- Walking
- Walking Upstairs
- Walking Downstairs
- Sitting
- Standing
- Laying

The dataset is high-dimensional because each sample has 561 input features. This motivates the use of PCA and SFFS to study whether classification performance can be maintained while reducing the number of features.

2.2 WISDM Dataset

The WISDM dataset consists of raw accelerometer data collected from users performing different physical activities. Unlike UCI HAR, which already provides engineered features, WISDM contains raw sensor readings along the x, y, and z axes. Therefore, feature extraction must be performed before applying classification algorithms.

In this project, the raw accelerometer data was divided into sliding windows. For each window, statistical features were extracted from each axis and from the signal magnitude. The extracted features include mean, standard deviation, minimum, maximum, median, interquartile range, and energy. This produced a lower-dimensional feature representation compared to UCI HAR.

The WISDM dataset is more challenging because raw sensor data is generally noisier and less structured than engineered features. It also contains a larger number of activity classes represented by letter labels. Therefore, it provides a useful comparison against the UCI HAR dataset.

2.3 Comparison of Datasets

Table 1 summarizes the two datasets used in this project.

Table 1: Summary of datasets used in the project

Dataset	Data Type	Number of Features	Activity Classes
UCI HAR	Engineered sensor features	561	6
WISDM	Raw accelerometer-derived features	28	18

The UCI HAR dataset is expected to produce higher classification accuracy because it uses carefully engineered features. The WISDM dataset is expected to be harder because it is based on raw accelerometer signals and contains more activity classes.

3 Description of Analysis Conducted

3.1 Data Preprocessing

Two normalization techniques were applied: z-score normalization and min–max normalization. Z-score normalization transforms each feature so that it has zero mean and unit variance. Min–max normalization rescales each feature into a fixed range, typically between 0 and 1.

Normalization is important because classifiers such as SVM and kNN are sensitive to feature scale. If features have very different numerical ranges, features with larger magnitudes may dominate the distance or decision function. By normalizing the data, each feature contributes more fairly to the classification process.

3.2 Classification Methods

The following classifiers were evaluated.

3.2.1 k-Nearest Neighbors

The k-Nearest Neighbors classifier assigns a test sample to the most common class among its nearest training samples. It is simple and effective, but it is sensitive to feature scaling and local data structure.

3.2.2 Support Vector Machine

Support Vector Machines attempt to find a separating decision boundary that maximizes the margin between classes. In this project, an RBF-kernel SVM was used because the activity classes are not necessarily linearly separable.

3.2.3 Random Forest

Random Forest is an ensemble method that combines multiple decision trees. It is robust to noise and can model nonlinear relationships between features. It also provides stable performance across different train-test splits.

3.2.4 Gaussian Naive Bayes

Gaussian Naive Bayes is a probabilistic classifier that assumes features are conditionally independent given the class. Although this assumption is often unrealistic for sensor data, the method is computationally efficient.

3.2.5 Multivariate Gaussian MAP Classifier

A Multivariate Gaussian classifier was implemented using the Maximum A Posteriori (MAP) decision rule. For each class, the class-conditional density was modeled using a multivariate Gaussian distribution. The class with the highest posterior probability was selected.

3.2.6 Parzen/KDE Classifier

A non-parametric Parzen window classifier was also implemented using kernel density estimation. Unlike Gaussian parametric methods, KDE does not assume that each class follows a single Gaussian distribution. However, it is computationally more expensive.

3.3 Principal Component Analysis

Principal Component Analysis (PCA) was applied for dimensionality reduction. PCA transforms the original features into a new set of orthogonal components ordered by the amount of variance they explain. In this project, PCA was used to retain 95% of the variance.

For the UCI HAR dataset, PCA reduced the dimensionality from 561 features to 104 principal components while retaining approximately 95.09% of the variance. For the WISDM dataset, PCA reduced the dimensionality from 28 features to 8 principal components while retaining approximately 95.07% of the variance.

3.4 Sequential Forward Floating Selection

Sequential Forward Floating Selection (SFFS) was used for feature selection. Unlike PCA, which creates new transformed features, SFFS selects a subset of the original features. The method begins with an empty feature set and adds features that improve validation performance. It can also remove previously selected features if doing so improves performance.

For UCI HAR, 12 features were selected. For WISDM, 10 features were selected. This allows the project to evaluate whether a small subset of informative features can achieve performance comparable to the full feature set.

3.5 Evaluation Metrics

The models were evaluated using classification accuracy, error rate, confusion matrices, and stability analysis. Accuracy measures the fraction of correctly classified samples. Error rate is computed as $1 - \text{accuracy}$. Confusion matrices were used to analyze which activity classes were most frequently confused.

To evaluate stability, the experiments were repeated over multiple train-test splits. The mean and variance of accuracy and error were computed. A low variance indicates that a classifier performs consistently across different partitions of the data.

4 Presentation of Results

4.1 Normalization Results

Table 2 shows the normalization comparison for the UCI HAR dataset. Z-score normalization produced the best SVM accuracy of 0.9918. kNN and Random Forest performed similarly under both normalization methods. Gaussian Naive Bayes produced lower accuracy because the Gaussian and independence assumptions do not fully match the structure of the sensor features.

Table 2: Normalization comparison on UCI HAR

Normalization	Classifier	Accuracy	Error Rate
Z-score	kNN	0.9658	0.0342
Z-score	SVM	0.9918	0.0082
Z-score	Random Forest	0.9744	0.0256
Z-score	Gaussian Naive Bayes	0.7468	0.2532
Min–max	kNN	0.9654	0.0346
Min–max	SVM	0.9872	0.0128
Min–max	Random Forest	0.9744	0.0256
Min–max	Gaussian Naive Bayes	0.7468	0.2532

Table 3 shows the same comparison for the WISDM dataset. Random Forest achieved the highest accuracy, followed by kNN. SVM performed much worse on WISDM than on UCI HAR, likely because WISDM contains raw-derived features and more classes.

Table 3: Normalization comparison on WISDM

Normalization	Classifier	Accuracy	Error Rate
Z-score	kNN	0.8587	0.1413
Z-score	SVM	0.5272	0.4728
Z-score	Random Forest	0.8818	0.1182
Z-score	Gaussian Naive Bayes	0.2447	0.7553
Min–max	kNN	0.8517	0.1483
Min–max	SVM	0.4004	0.5996
Min–max	Random Forest	0.8816	0.1184
Min–max	Gaussian Naive Bayes	0.2447	0.7553

4.2 PCA Results

PCA was applied while retaining 95% of the variance. Table 4 summarizes the results. For UCI HAR, PCA reduced the number of features from 561 to 104. The SVM accuracy after PCA was 0.9786, which is slightly lower than the full-feature SVM accuracy but still very strong. For WISDM, PCA reduced the number of features from 28 to 8, but accuracy decreased slightly for most models.

Table 4: PCA results with 95% variance retained

Dataset	Classifier	PCA Components	Accuracy	Error Rate
UCI HAR	kNN	104	0.9604	0.0396
UCI HAR	SVM	104	0.9786	0.0214
UCI HAR	Random Forest	104	0.9371	0.0629
UCI HAR	Gaussian Naive Bayes	104	0.8066	0.1934
WISDM	kNN	8	0.8315	0.1685
WISDM	SVM	8	0.4767	0.5233
WISDM	Random Forest	8	0.8478	0.1522
WISDM	Gaussian Naive Bayes	8	0.2337	0.7663

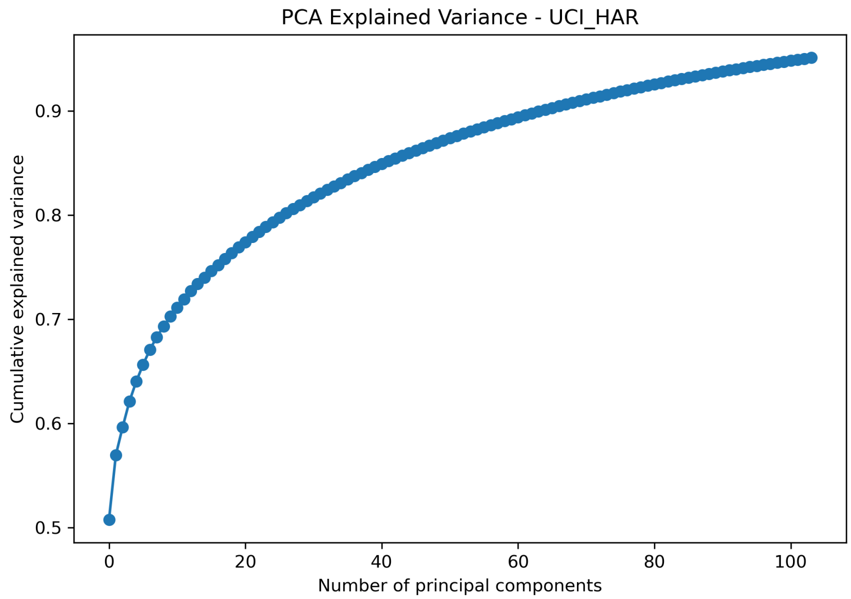

Figure 1: Cumulative explained variance using PCA on the UCI HAR dataset.

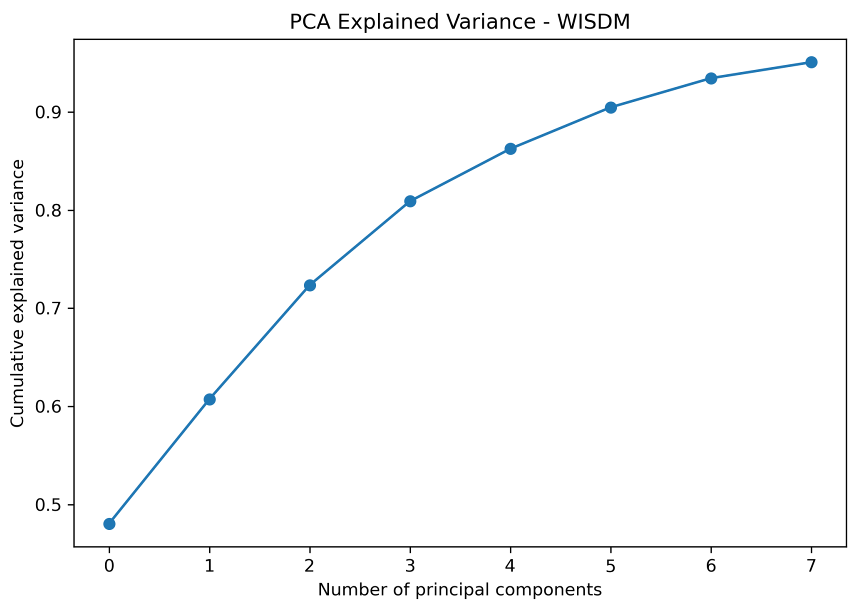

Figure 2: Cumulative explained variance using PCA on the WISDM dataset.

4.3 SFFS Results

Table 5 shows the results after applying SFFS. For UCI HAR, only 12 selected features were used. The kNN classifier achieved 0.9763 accuracy, which is even higher than its full-feature accuracy. Random Forest also remained strong with 0.9728 accuracy. This shows that many of the 561 UCI HAR features are redundant.

For WISDM, 10 selected features were used. Random Forest achieved 0.8742 accuracy and kNN achieved 0.8625 accuracy. These results are close to the full-feature WISDM results, showing that a small set of statistical features captures much of the useful activity information.

Table 5: SFFS feature selection results

Dataset	Classifier	Selected Features	Accuracy	Error Rate
UCI HAR	kNN	12	0.9763	0.0237
UCI HAR	SVM	12	0.9348	0.0652
UCI HAR	Random Forest	12	0.9728	0.0272
UCI HAR	Gaussian Naive Bayes	12	0.8435	0.1565
WISDM	kNN	10	0.8625	0.1375
WISDM	SVM	10	0.4821	0.5179
WISDM	Random Forest	10	0.8742	0.1258
WISDM	Gaussian Naive Bayes	10	0.2507	0.7493

4.4 Bayesian Classifier Results

Table 6 shows the Bayesian classifier results. On UCI HAR, the Multivariate Gaussian MAP classifier achieved 0.8959 accuracy and the Parzen/KDE classifier achieved 0.8893 accuracy. Gaussian Naive Bayes was lower at 0.8089 when using PCA components for the Bayesian experiment.

On WISDM, Bayesian classifiers performed much worse than kNN and Random Forest. The Parzen/KDE classifier achieved the best Bayesian result on WISDM with 0.3921 accuracy. This indicates that the raw-derived WISDM features are harder to model using simple probabilistic assumptions.

Table 6: Bayesian classifier results

Dataset	Classifier	PCA Components Used	Accuracy	Error Rate
UCI HAR	Multivariate Gaussian MAP	20	0.8959	0.1041
UCI HAR	Gaussian Naive Bayes	20	0.8089	0.1911
UCI HAR	Parzen/KDE MAP	20	0.8893	0.1107
WISDM	Multivariate Gaussian MAP	10	0.2841	0.7159
WISDM	Gaussian Naive Bayes	10	0.2585	0.7415
WISDM	Parzen/KDE MAP	10	0.3921	0.6079

4.5 Stability Analysis Results

Table 7 shows the stability analysis for UCI HAR over multiple train-test splits. SVM achieved the highest mean accuracy of 0.9859 and had very low variance. Random Forest also showed strong stability with mean accuracy 0.9776. Gaussian Naive Bayes had the lowest mean accuracy and highest variance.

Table 7: Stability analysis on UCI HAR

Classifier	Mean Accuracy	Variance Accuracy	Mean Error	Variance Error
Gaussian Naive Bayes	0.7505	0.001476	0.2495	0.001476
Random Forest	0.9776	0.000012	0.0224	0.000012
SVM	0.9859	0.000008	0.0141	0.000008
kNN	0.9604	0.000019	0.0396	0.000019

4.6 Confusion Matrix Results

Confusion matrices were generated for the main classifiers. Figure 3 shows the confusion matrix for SVM on UCI HAR. SVM achieved the best overall performance, with very few misclassifications. The most likely confusions occur between sitting and standing, which is expected because both are static activities with similar sensor patterns.

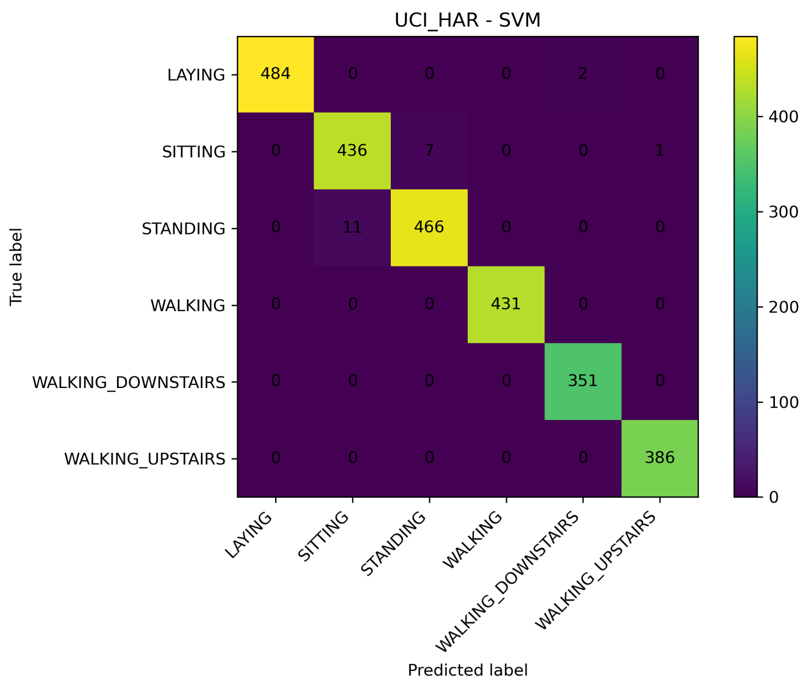

Figure 3: Confusion matrix for SVM on the UCI HAR dataset.

Figure 4 shows the Random Forest confusion matrix for UCI HAR. Random Forest also performed strongly, although SVM achieved slightly higher accuracy.

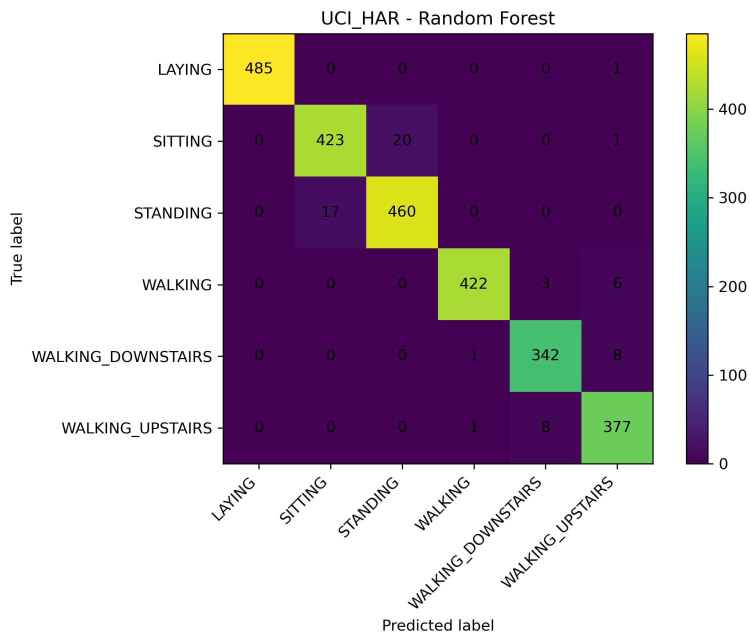

Figure 4: Confusion matrix for Random Forest on the UCI HAR dataset.

Figure 5 shows the Random Forest confusion matrix for WISDM. Random Forest was the best-performing model on WISDM. However, the confusion matrix shows more misclassifications compared to UCI HAR, which reflects the difficulty of classifying raw-derived sensor features across a larger number of activity classes.

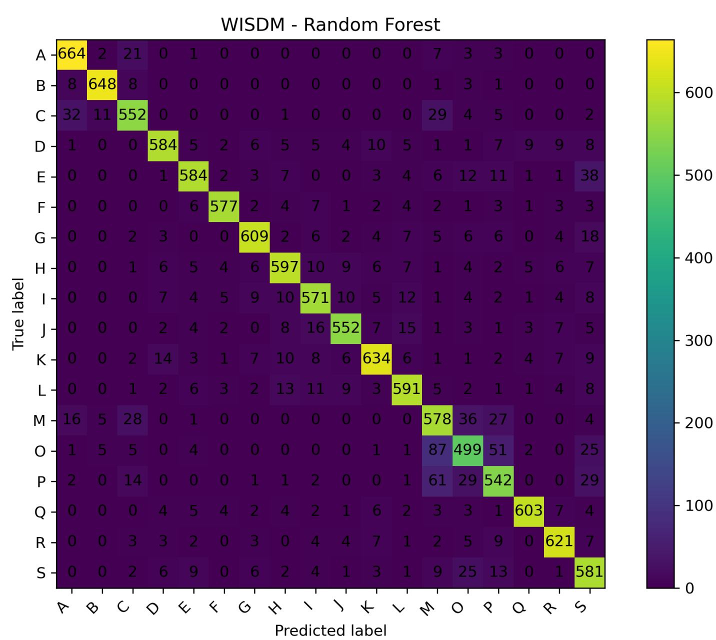

Figure 5: Confusion matrix for Random Forest on the WISDM dataset.

5 Analysis of Results

5.1 Comparison Between Datasets

The results clearly show that the UCI HAR dataset is easier to classify than the WISDM dataset. The highest accuracy on UCI HAR was achieved by SVM with z-score normalization (0.9918). In contrast, the best WISDM result was achieved by Random Forest (0.8818).

This difference is expected. The UCI HAR dataset contains engineered features that already capture important signal characteristics. On the other hand, WISDM relies on raw sensor-derived features, which are noisier and less structured. As a result, classification becomes more difficult.

5.2 Effect of Normalization

Normalization affects classifiers differently.

SVM benefits significantly from z-score normalization. On UCI HAR, SVM accuracy decreases from 0.9918 (z-score) to 0.9872 (min–max). On WISDM, the drop is more substantial, from 0.5272 to 0.4004. This confirms that SVM is highly sensitive to feature scaling.

In contrast, Random Forest is largely unaffected by normalization. This is because tree-based models do not rely on distance metrics or feature magnitude in the same way as SVM or kNN.

5.3 Impact of PCA

PCA reduces dimensionality while preserving most of the variance.

For UCI HAR, PCA reduces the feature space from 561 to 104 components. The SVM accuracy decreases slightly from 0.9918 to 0.9786, which is a small trade-off for a large reduction in dimensionality.

For WISDM, PCA reduces the feature space from 28 to 8 components. However, the drop in accuracy is more noticeable. This suggests that PCA removes some useful discriminative information in this dataset.

5.4 Impact of Feature Selection (SFFS)

SFFS identifies a small subset of important features.

For UCI HAR, kNN achieves 0.9763 accuracy using only 12 features, which is higher than its full-feature accuracy. This indicates that many features in the dataset are redundant.

For WISDM, Random Forest achieves 0.8742 accuracy using 10 features, which is close to the full-feature result. This shows that a compact feature set can still capture most of the useful information.

5.5 Bayesian vs Machine Learning Models

Machine learning models outperform Bayesian classifiers in most cases.

On UCI HAR:

- SVM: 0.9918 (best)
- Random Forest: 0.9744
- Multivariate Gaussian: 0.8959
- Parzen/KDE: 0.8893

On WISDM:

- Random Forest: 0.8818 (best)
- kNN: 0.8587
- Parzen/KDE: 0.3921
- Gaussian Naive Bayes: 0.2585

Bayesian models perform worse because they rely on assumptions (Gaussian distribution or independence) that do not hold well for real sensor data.

5.6 Confusion Matrix Insights

Confusion matrices reveal which classes are hardest to distinguish.

On UCI HAR:

- Sitting and Standing are frequently confused
- Walking-related activities also show some overlap
- Laying is classified with very high accuracy

These results are expected because similar physical movements produce similar sensor signals.

5.7 Stability Analysis

Stability analysis evaluates how consistent the models are.

SVM and Random Forest show:

- High mean accuracy
- Very low variance

This indicates that their performance is stable across different train-test splits.

Gaussian Naive Bayes shows:

- Lower mean accuracy
- Higher variance

6 Summary and Conclusions

This project studied human activity recognition using smartphone sensor data. Two datasets were used: UCI HAR and WISDM. These datasets allowed a comparison between engineered features and raw sensor-derived features.

Several classifiers were evaluated, including kNN, SVM, Random Forest, and Bayesian models. Additional techniques such as normalization, PCA, and SFFS were also applied.

The key findings are summarized below:

- SVM achieved the best performance on UCI HAR (0.9918 accuracy)
- Random Forest achieved the best performance on WISDM (0.8818 accuracy)
- Z-score normalization performed better than min–max for SVM
- PCA reduced dimensionality significantly with only a small loss in accuracy
- SFFS showed that a small number of features can achieve strong performance
- Bayesian classifiers performed worse due to strong distributional assumptions
- UCI HAR is easier to classify due to engineered features
- WISDM is more challenging due to raw sensor data and more activity classes

Overall, the results demonstrate that feature representation plays a critical role in classification performance. Engineered features lead to higher accuracy, while raw data requires more robust models and careful preprocessing.

Future work could explore deep learning methods, additional sensor modalities, and improved feature extraction techniques for raw sensor data.